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1 Addendum: projective completion and the square-root causal residual

Version 4 established the two-action causal Shannon-floor theorem

$$C_\infty^{\text{causal}}(P, Q) = \max\{H(P), H(Q)\},$$

together with the finite-horizon estimate

$$C_n^{\text{causal}} \leq nh_{\max} + O(n^{2/3}).$$

It left open two logically distinct questions:

1. whether the finite-horizon optima can be embedded into one projectively consistent infinite causal substrate with the same asymptotic entropy rate;
2. whether the finite-horizon causal residual

$$E_n := C_n^{\text{causal}} - nh$$

in the equal-entropy case is bounded, logarithmic, of square-root order, or obeys some other sublinear law.

Both questions admit sharper answers.

The first is resolved abstractly by concatenating increasingly long finite causal trees. The second is resolved by combining an exact one-sided interval coupling with the anti-diagonal causal lift.

For finite equal-entropy pairs with unequal varentropy,

$$\boxed{C_n^{\text{causal}}(P, Q) = nh + \Theta(\sqrt{n}).}$$

In particular, the crash-test pair of Version 4 has a genuinely diverging square-root boundary bill even though its asymptotic rate is exactly the Shannon floor.

The exact coefficient multiplying $\sqrt{n}$ is not determined here.

1.1 One infinite substrate from finite causal blocks

Recall that

$$C_n^{\text{causal}}$$

denotes the minimum entropy of an exact horizon- n causal action tree, and that

$$C_\infty^{\text{causal}} = \lim_{n \rightarrow \infty} \frac{C_n^{\text{causal}}}{n} = \inf_{n \geq 1} \frac{C_n^{\text{causal}}}{n}$$

exists by subadditivity.

For an infinite random causal action tree $\mathcal{T}$, write $\mathcal{T}_{\leq n}$ for its restriction to depth n .

Theorem 1.1 (Projective completion theorem). *For every finite controlled memoryless channel, there exists one projectively consistent infinite exact causal action tree $\mathcal{T}$ such that*

$$\lim_{n \rightarrow \infty} \frac{H(\mathcal{T}_{\leq n})}{n} = C_{\infty}^{\text{causal}}.$$

Consequently,

$$\inf_{\mathcal{T}} \limsup_{n \rightarrow \infty} \frac{H(\mathcal{T}_{\leq n})}{n} = C_{\infty}^{\text{causal}},$$

where the infimum is over projectively consistent infinite exact causal substrates.

Proof. The lower bound is immediate. For every infinite exact causal tree $\mathcal{T}$, its depth- n restriction is an admissible horizon- n causal tree. Hence

$$H(\mathcal{T}_{\leq n}) \geq C_n^{\text{causal}},$$

and therefore

$$\liminf_{n \rightarrow \infty} \frac{H(\mathcal{T}_{\leq n})}{n} \geq C_{\infty}^{\text{causal}}.$$

For the reverse direction, let

$$B_j$$

be an exact causal action tree of depth j satisfying

$$H(B_j) \leq C_j^{\text{causal}} + \varepsilon_j, \quad \varepsilon_j := 2^{-j}.$$

Take the family $(B_j)_{j \geq 1}$ independently.

Partition absolute time into consecutive blocks of lengths

$$1, 2, 3, \dots$$

Write

$$N_j := 1 + 2 + \dots + j = \frac{j(j+1)}{2}.$$

During block j , use the same random local tree B_j after every history entering that block. Equivalently, the response inside the block depends only on the action suffix since the most recent block boundary and on B_j , but not on the earlier history.

Every deterministic action word therefore traverses one valid path inside each block. The output law inside block j is the required product channel law because B_j is exact, while different blocks are independent. Concatenation consequently gives the exact product law along every deterministic infinite action word. By the action-tree characterization, the resulting infinite tree is exact under every adaptive nonanticipative policy.

Now let

$$N_{j-1} < n \leq N_j.$$

The restriction $\mathcal{T}_{\leq n}$ is a deterministic function of

$$B_1, \dots, B_j,$$

so

$$H(\mathcal{T}_{\leq n}) \leq \sum_{i < j} (C_i^{\text{causal}} + \varepsilon_i) + C_j^{\text{causal}} + \varepsilon_j.$$

Subadditivity gives

$$C_j^{\text{causal}} \leq j C_1^{\text{causal}},$$

and hence

$$\frac{C_j^{\text{causal}}}{N_{j-1}} \rightarrow 0.$$

Moreover,

$$\frac{\sum_{i < j} C_i^{\text{causal}}}{N_{j-1}} = \frac{\sum_{i < j} i (C_i^{\text{causal}}/i)}{\sum_{i < j} i} \rightarrow C_\infty^{\text{causal}}$$

by weighted Cesàro convergence. The summable errors ε_i contribute $o(N_j)$.

Therefore

$$\limsup_{n \rightarrow \infty} \frac{H(\mathcal{T}_{\leq n})}{n} \leq C_\infty^{\text{causal}}.$$

Together with the universal lower bound, this proves

$$\lim_{n \rightarrow \infty} \frac{H(\mathcal{T}_{\leq n})}{n} = C_\infty^{\text{causal}}.$$

□

Corollary 1.2 (Infinite two-action Shannon floor). *For every finite two-action memoryless channel, there exists one projectively consistent infinite exact causal substrate whose entropy rate is*

$$\max\{H(P), H(Q)\}.$$

For the crash-test pair,

$$\boxed{\lim_{n \rightarrow \infty} \frac{H(\mathcal{T}_{\leq n})}{n} = 1.5.}$$

Remark 1.3. *The construction in Theorem 1.1 is generally nonstationary: the block lengths increase with absolute time. Thus projective consistency is settled in the sense of Definition 24.1, but a stronger requirement of stationarity or shift-invariance would define a different problem.*

1.2 Exact one-sided interval lifting

Let P and Q be finite probability distributions after deletion of zero-probability symbols. For a word $x^t = (x_1, \dots, x_t)$ define its accumulated surprisal

$$I_P(x^t) := -\log_2 P^{\otimes t}(x^t) = -\sum_{s=1}^t \log_2 P(x_s),$$

and analogously define $I_Q(w^t)$.

We now construct a coupling of

$$X^n \sim P^{\otimes n}, \quad W^n \sim Q^{\otimes n}$$

which is causal from W to X and whose conditional entropy is controlled by the maximal prefix information deficit.

Lemma 1.4 (Exact causal interval lifting). *For every $n \geq 1$ there exists a coupling of*

$$X^n \sim P^{\otimes n}, \quad W^n \sim Q^{\otimes n}$$

such that, for every $t \leq n$,

$$\text{Law}(X_{1:t} \mid W_{1:n}) = \text{Law}(X_{1:t} \mid W_{1:t}). \quad (30.1)$$

Moreover,

$$H(X^n | W^n) \leq \mathbb{E} \max_{0 \leq t \leq n} \{I_P(X_{1:t}) - I_Q(W_{1:t})\} + \log_2(2n) + c_{\text{int}}, \quad (30.2)$$

where one may take

$$c_{\text{int}} = 1 + \frac{1}{\ln 2}.$$

Proof. Use the standard arithmetic partition of $[0, 1)$ associated with P . For every word x^t , let

$$I(x^t) \subset [0, 1)$$

be its depth- t cylinder, so that

$$|I(x^t)| = P^{\otimes t}(x^t).$$

Take an i.i.d. source

$$W_1, W_2, \dots \sim Q$$

and, independently, an infinite sequence of fair bits

$$R_1, R_2, \dots$$

Maintain a nested interval J . Initially

$$J = [0, 1).$$

At step t , first partition the current interval into consecutive pieces in the proportions $Q(w)$ and select the piece indexed by W_t . Then repeatedly bisect the selected interval and choose the left or right half according to fresh fair bits until the resulting interval is contained in one depth- t P -cylinder. Declare that cylinder to be $I(X_{1:t})$.

Every elementary split chooses a child interval with probability equal to its relative length. Hence, inductively, the probability of reaching any current interval equals its Lebesgue length. The nested intervals therefore determine a uniform point

$$U \in [0, 1)$$

outside the countable set of arithmetic-partition boundaries with probability one.

Consequently

$$X_{1:t} = x^t \iff U \in I(x^t),$$

and hence

$$\Pr(X_{1:t} = x^t) = |I(x^t)| = P^{\otimes t}(x^t).$$

Thus

$$X^n \sim P^{\otimes n}.$$

The construction of $X_{1:t}$ uses only

$$W_{1:t}$$

and fair bits already consumed by time t , and therefore satisfies the causality condition (??).

Let B_t denote the total number of fair bits consumed by time t . After reading $W_{1:t}$ and B_t fair bits, the current interval has length

$$|J_t| = Q^{\otimes t}(W_{1:t}) 2^{-B_t}. \quad (30.3)$$

Let

$$\delta_t$$

be the distance from U to the nearer boundary of the target cylinder $I(X_{1:t})$. Once

$$|J_t| < \delta_t,$$

every interval containing U is necessarily contained in $I(X_{1:t})$. Hence

$$B_n \leq 1 + \max_{1 \leq t \leq n} \left[\log_2 \frac{Q^{\otimes t}(W_{1:t})}{\delta_t} \right]_+. \quad (30.4)$$

Write

$$\delta_t = P^{\otimes t}(X_{1:t}) D_t, \quad 0 < D_t \leq \frac{1}{2}.$$

Then

$$\log_2 \frac{Q^{\otimes t}(W_{1:t})}{\delta_t} = I_P(X_{1:t}) - I_Q(W_{1:t}) + \log_2 \frac{1}{D_t}.$$

Since the $t = 0$ information deficit is zero,

$$B_n \leq 1 + \max_{0 \leq t \leq n} \{I_P(X_{1:t}) - I_Q(W_{1:t})\} + G_n, \quad (30.5)$$

where

$$G_n := \max_{1 \leq t \leq n} \log_2 \frac{1}{D_t}.$$

Conditioned on $X_{1:t}$, the normalized location of U inside its cylinder is uniform on $[0, 1)$. Therefore

$$\Pr(D_t \leq \varepsilon \mid X_{1:t}) = 2\varepsilon, \quad 0 \leq \varepsilon \leq \frac{1}{2}.$$

A union bound gives

$$\Pr(G_n > s) \leq \min\{1, 2n 2^{-s}\}. \quad (30.6)$$

Integrating the tail,

$$\mathbb{E}G_n \leq \log_2(2n) + \frac{1}{\ln 2}. \quad (30.7)$$

It remains to relate consumed fair bits to conditional entropy. For fixed $W^n = w^n$, the possible terminal fair-bit transcripts form a prefix-free stopping set. If r is a terminal transcript, then

$$\Pr(R = r \mid W^n = w^n) = 2^{-|r|}.$$

Hence

$$H(R \mid W^n = w^n) = \sum_r 2^{-|r|} |r| = \mathbb{E}(B_n \mid W^n = w^n).$$

Since X^n is a deterministic function of (W^n, R) ,

$$H(X^n \mid W^n) \leq H(R \mid W^n) = \mathbb{E}B_n.$$

Combining this with (??)–(??) proves (??). $\square$

Remark 1.5 (No square-root discretization tax). *The interval construction is exact at the atomic level. Its additional discretization overhead is only*

$$O(\log n) = o(\sqrt{n}).$$

Thus exact atom splitting does not itself create a new second-order $\sqrt{n}$ contribution.

1.3 Anti-diagonal lifting of a causal stream coupling

The previous lemma constructs only two coupled streams. The next observation turns such a one-sided causal coupling into one complete causal response tree.

Proposition 1.6 (Stream-to-tree anti-diagonal lift). *Suppose*

$$X^n \sim P^{\otimes n}, \quad W^n \sim Q^{\otimes n}$$

satisfy

$$\text{Law}(X_{1:t} \mid W_{1:n}) = \text{Law}(X_{1:t} \mid W_{1:t}) \quad \forall t \leq n. \quad (30.8)$$

Define the reversed Q -stream

$$Z_s := W_{n-s+1}, \quad 1 \leq s \leq n.$$

Use the deterministic response rule

$$\begin{cases} \text{on the } r\text{-th occurrence of action } P, \text{ output } X_r, \\ \text{on the } s\text{-th occurrence of action } Q, \text{ output } Z_s. \end{cases} \quad (30.9)$$

Then (??) is an exact horizon- n causal realization. Its seed entropy satisfies

$$C_n^{\text{causal}}(P, Q) \leq H(X^n, W^n). \quad (30.10)$$

Proof. Fix a deterministic action word of length at most n . Suppose it contains p occurrences of P and q occurrences of Q . The realized outputs consist of

$$X_{1:p}$$

and

$$W_{n-q+1:n}.$$

Since

$$p + q \leq n,$$

we have

$$n - q + 1 \geq p + 1.$$

Thus the observed W -suffix is contained entirely in the future coordinates $W_{p+1:n}$.

By (??),

$$\Pr(X_{1:p} \in A \mid W_{1:n}) = \Pr(X_{1:p} \in A \mid W_{1:p}).$$

Since W is i.i.d., the suffix $W_{n-q+1:n}$ is independent of $W_{1:p}$. Consequently,

$$X_{1:p} \perp W_{n-q+1:n}.$$

Their marginal laws are respectively

$$P^{\otimes p} \quad \text{and} \quad Q^{\otimes q}.$$

Hence every deterministic action word has exactly the required memoryless product output law.

The action-tree characterization then implies exactness for every adaptive nonanticipative policy. The entire tree is a deterministic function of (X^n, W^n) , proving (??). $\square$

If

$$H(P) = H(Q) = h,$$

Proposition 1.6 gives

$$\begin{aligned} C_n^{\text{causal}}(P, Q) &\leq H(X^n, W^n) \\ &= nh + H(X^n \mid W^n). \end{aligned}$$

Therefore

$$E_n \leq H(X^n \mid W^n). \quad (30.11)$$

1.4 A square-root upper bound

Assume from now on that

$$H(P) = H(Q) = h.$$

Define the varentropies

$$V_P := \text{Var}(-\log_2 P(X)), \quad X \sim P,$$

and

$$V_Q := \text{Var}(-\log_2 Q(W)), \quad W \sim Q.$$

Write

$$\sigma_P := \sqrt{V_P}, \quad \sigma_Q := \sqrt{V_Q}.$$

Define the centered surprisal processes

$$M_t^P := I_P(X_{1:t}) - th, \quad M_t^Q := I_Q(W_{1:t}) - th. \quad (30.12)$$

Theorem 1.7 (Square-root causal upper bound). *For every finite equal-entropy pair P, Q ,*

$$\boxed{E_n \leq 2(\sigma_P + \sigma_Q)\sqrt{n} + \log_2(2n) + c_{\text{int}}.} \quad (30.13)$$

In particular,

$$\boxed{E_n = O(\sqrt{n}).} \quad (30.14)$$

More sharply,

$$\boxed{\limsup_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \leq \sqrt{\frac{2}{\pi}}(\sigma_P + \sigma_Q).} \quad (30.15)$$

Proof. By Lemma 1.4 and Proposition 1.6,

$$E_n \leq \mathbb{E} \max_{0 \leq t \leq n} (M_t^P - M_t^Q) + \log_2(2n) + c_{\text{int}}. \quad (30.16)$$

Pathwise,

$$\max_{t \leq n} (M_t^P - M_t^Q) \leq \max_{t \leq n} |M_t^P| + \max_{t \leq n} |M_t^Q|. \quad (30.17)$$

Under their respective marginal laws, M^P and M^Q are centered martingales with

$$\mathbb{E}(M_n^P)^2 = nV_P, \quad \mathbb{E}(M_n^Q)^2 = nV_Q.$$

Doob's L^2 maximal inequality therefore gives

$$\mathbb{E} \max_{t \leq n} |M_t^P| \leq 2\sigma_P \sqrt{n}$$

and

$$\mathbb{E} \max_{t \leq n} |M_t^Q| \leq 2\sigma_Q \sqrt{n}.$$

Substitution into (??) proves (??).

For the asymptotic refinement, Donsker's invariance principle gives

$$\left(\frac{M_{[nt]}^P}{\sqrt{n}} \right)_{0 \leq t \leq 1} \Longrightarrow (\sigma_P B_t)_{0 \leq t \leq 1},$$

and likewise for Q . The L^2 maximal bound above supplies uniform integrability, so

$$\frac{1}{\sqrt{n}} \mathbb{E} \max_{t \leq n} M_t^P \longrightarrow \sigma_P \mathbb{E} \sup_{0 \leq t \leq 1} B_t = \sigma_P \sqrt{\frac{2}{\pi}}.$$

Similarly,

$$\frac{1}{\sqrt{n}} \mathbb{E} \max_{t \leq n} (-M_t^Q) \longrightarrow \sigma_Q \sqrt{\frac{2}{\pi}}.$$

Using

$$\max_t (M_t^P - M_t^Q) \leq \max_t M_t^P + \max_t (-M_t^Q)$$

and noting that $\log n = o(\sqrt{n})$ proves (??). $\square$

1.5 A square-root converse from two deterministic policies

The matching order lower bound does not require the full family of adaptive policies. It is already forced by the two constant policies.

Theorem 1.8 (Varentropy-mismatch converse). *For every finite equal-entropy pair P, Q ,*

$$E_n \geq \frac{1}{2} W_1 (\text{Law}(M_n^P), \text{Law}(M_n^Q)). \quad (30.18)$$

Consequently,

$$\boxed{\liminf_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \geq \frac{|\sigma_P - \sigma_Q|}{\sqrt{2\pi}}.} \quad (30.19)$$

Proof. Let S_n be any exact horizon- n causal seed. Apply first the deterministic policy that always chooses P , and then the deterministic policy that always chooses Q . The corresponding output words

$$X^n = f(S_n), \quad Z^n = g(S_n)$$

are deterministic functions of the same seed and satisfy

$$X^n \sim P^{\otimes n}, \quad Z^n \sim Q^{\otimes n}.$$

Hence

$$H(S_n) \geq H(X^n, Z^n). \quad (30.20)$$

Let $\Gamma(x, z)$ denote their joint law. Every joint atom satisfies

$$\Gamma(x, z) \leq P^{\otimes n}(x)$$

and

$$\Gamma(x, z) \leq Q^{\otimes n}(z).$$

Therefore

$$-\log_2 \Gamma(x, z) \geq \max\{I_P(x), I_Q(z)\}.$$

Averaging,

$$H(X^n, Z^n) \geq \mathbb{E} \max\{I_P(X^n), I_Q(Z^n)\}. \quad (30.21)$$

Since both entropies equal nh ,

$$I_P(X^n) = nh + M_n^P, \quad I_Q(Z^n) = nh + M_n^Q,$$

and

$$\mathbb{E} M_n^P = \mathbb{E} M_n^Q = 0.$$

Thus

$$\begin{aligned} H(X^n, Z^n) &\geq nh + \frac{1}{2} \mathbb{E} |M_n^P - M_n^Q| \\ &\geq nh + \frac{1}{2} W_1 (\text{Law}(M_n^P), \text{Law}(M_n^Q)). \end{aligned} \quad (30.22)$$

Minimizing over exact causal seeds proves (??).

The central limit theorem gives

$$\frac{M_n^P}{\sqrt{n}} \Rightarrow N(0, V_P), \quad \frac{M_n^Q}{\sqrt{n}} \Rightarrow N(0, V_Q).$$

Finite second moments imply uniform integrability, hence convergence also holds in W_1 . Therefore

$$\frac{1}{\sqrt{n}} W_1(\text{Law}(M_n^P), \text{Law}(M_n^Q)) \rightarrow W_1(N(0, V_P), N(0, V_Q)).$$

For centered one-dimensional Gaussians, monotone quantile coupling gives

$$W_1(N(0, V_P), N(0, V_Q)) = \sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q|.$$

Combining this identity with (??) proves (??). □

1.6 The square-root residual theorem

Combining the previous two estimates gives the finite-horizon law.

Theorem 1.9 (Square-root causal residual). *Let P, Q be finite distributions satisfying*

$$H(P) = H(Q) = h.$$

If

$$V_P \neq V_Q,$$

then

$$\boxed{C_n^{\text{causal}}(P, Q) = nh + \Theta(\sqrt{n}).} \tag{30.23}$$

Equivalently,

$$\boxed{E_n = \Theta(\sqrt{n}).} \tag{30.24}$$

More explicitly,

$$\frac{|\sigma_P - \sigma_Q|}{\sqrt{2\pi}} \leq \liminf_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \leq \limsup_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \leq \sqrt{\frac{2}{\pi}} (\sigma_P + \sigma_Q). \tag{30.25}$$

Remark 1.10. *The hypothesis $V_P \neq V_Q$ is used only for the positive square-root lower bound. When $V_P = V_Q$, the present argument still proves*

$$E_n = O(\sqrt{n}),$$

but does not exclude a smaller order.

1.7 The crash-test pair

For the crash-test pair of Version 4,

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right), \quad Q = (q, q, 1 - 2q),$$

where q is chosen so that

$$H(P) = H(Q) = 1.5.$$

For P ,

$$V_P = \frac{1}{4}, \quad \sigma_P = \frac{1}{2}.$$

For Q ,

$$\sigma_Q = 0.456450775263963 \dots$$

Hence

$$V_P \neq V_Q.$$

Theorem 1.9 therefore gives

$$\boxed{C_n^{\text{causal}}(P, Q) = 1.5n + \Theta(\sqrt{n}).} \quad (30.26)$$

In particular,

$$\boxed{E_n \longrightarrow \infty, \quad \frac{E_n}{n} \longrightarrow 0.} \quad (30.27)$$

Thus the causal Shannon floor is approached in rate but not within a bounded additive amount. The finite-horizon boundary bill diverges, and its exact order is square root.

The simple two-policy converse yields numerically

$$\liminf_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \geq 0.017373627025909 \dots, \quad (30.28)$$

while the constructive upper bound gives

$$\limsup_{n \rightarrow \infty} \frac{E_n}{\sqrt{n}} \leq 0.763137306751047 \dots \quad (30.29)$$

The order is therefore determined rigorously even though the exact second-order coefficient remains open.

1.8 Updated status of the open problems

The preceding results change the status of several questions listed in Section 27.

Finite causal residual. For the crash-test pair, Open Problem 27.3 is resolved:

$$\boxed{E_n = \Theta(\sqrt{n}).}$$

Thus the residual is neither bounded nor logarithmic.

Projectively consistent infinite substrate. Open Problem 27.5 is resolved under Definition 24.1: there exists one projectively consistent infinite exact causal tree whose prefix entropy rate equals

$$C_\infty^{\text{causal}}.$$

For the crash-test pair this rate is exactly 1.5.

Sharper anti-diagonal coding. For equal-entropy two-action channels, the $O(n^{2/3})$ estimate of Version 4 improves to

$$E_n = O(\sqrt{n}).$$

When the action varentropies differ, Theorem 1.8 shows that this order cannot be improved:

$$E_n = \Omega(\sqrt{n}).$$

The natural remaining second-order problem is therefore no longer the growth exponent but the coefficient.

Open Problem 1.11 (Exact causal second-order constant). *For a finite equal-entropy pair with $V_P \neq V_Q$, determine whether the limit*

$$\kappa(P, Q) := \lim_{n \rightarrow \infty} \frac{C_n^{\text{causal}}(P, Q) - nh}{\sqrt{n}} \quad (30.30)$$

exists, and if so determine it exactly.

For the crash-test pair, the continuous adaptive-profile calculation singles out the candidate

$$\sqrt{\frac{2}{\pi}} |\sigma_P - \sigma_Q| = 0.034747254051818\dots, \quad (30.31)$$

but the equality

$$\kappa(P, Q) = 0.034747254051818\dots$$

is not claimed here. A matching discrete causal coupling at the sharp second-order constant would be required to establish it.

1.9 Revised causal picture

For the crash-test pair, the causal side of the resource hierarchy can now be stated more sharply:

$$\boxed{\begin{aligned} C_n^{\text{causal}} &= 1.5n + \Theta(\sqrt{n}), \\ C_\infty^{\text{causal}} &= 1.5, \\ \exists \mathcal{T}_\infty \quad \frac{H((\mathcal{T}_\infty)_{\leq n})}{n} &\longrightarrow 1.5. \end{aligned}} \quad (30.32)$$

The causal excess therefore has a boundary-layer character. It is unavoidable at every finite horizon and in fact diverges, but it is only of diffusive order and disappears after normalization by the horizon.

The distinction exposed in Version 4 becomes correspondingly sharper:

A history-indexed causal substrate can reach the observable Shannon floor with one projectively consistent infinite realization. Nevertheless, when the available actions have equal entropy but different information variance, every finite exact realization pays a nonzero boundary bill of order $\sqrt{n}$.

The asymptotic Shannon floor and the finite counterfactual surcharge are therefore not contradictory. They occupy different asymptotic orders.